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ACTION EFFECTS ON DISTANCE PERCEPTION: A META-ANALYSIS

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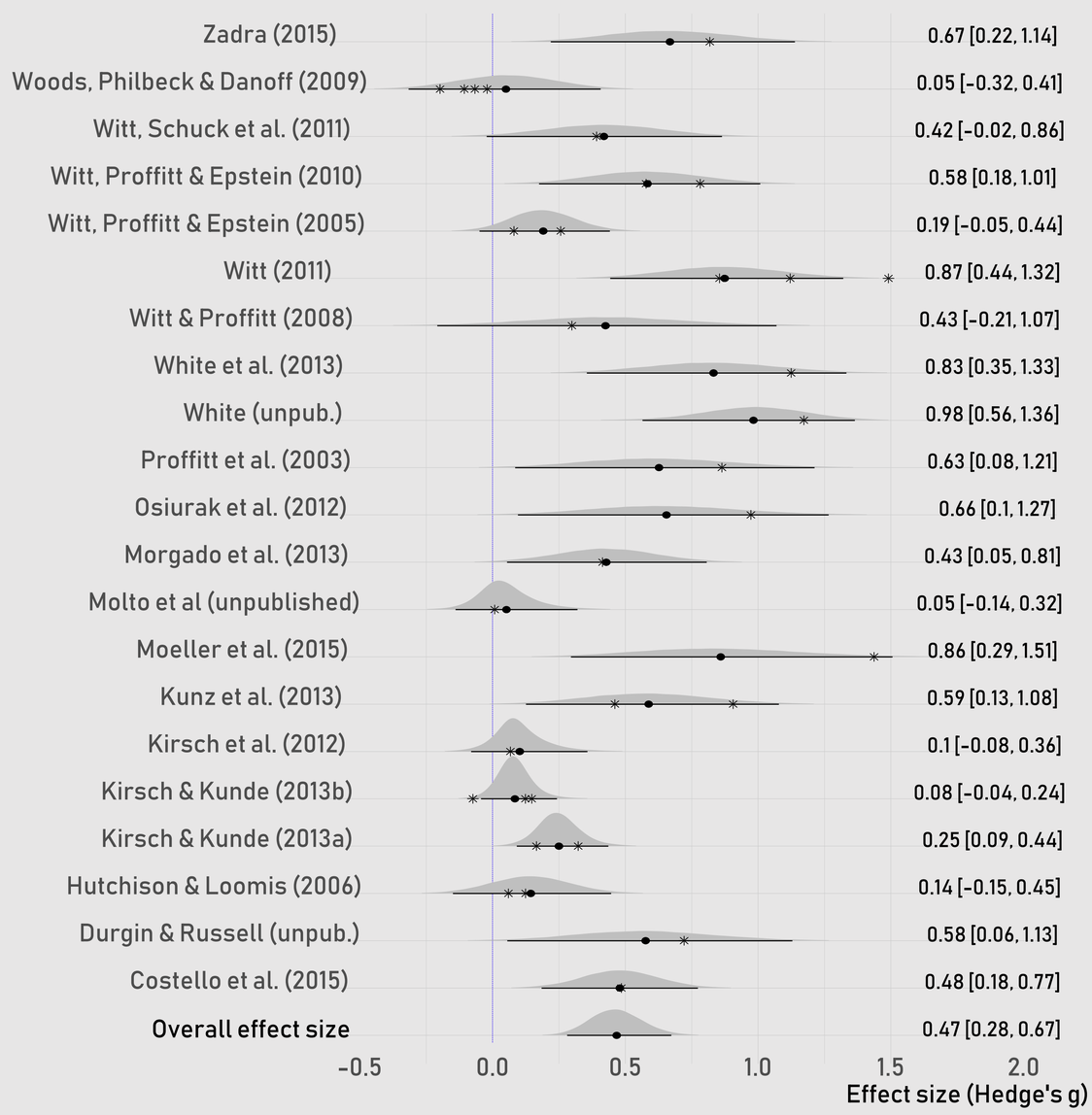
BACKGROUND

According to Action Constraint Theories (e.g., Witt, 2011; Morgado & Palluel-Germain, 2015), people perceive distances according to the variables defining their action abilities. For instance, participants estimated a target to be closer when they could use a tool to reach it more easily (i.e., low Action Constraint), than when they could not (i.e., high Action Constraint; Witt, 2011). Although many studies supported these theories, the existence and magnitude of such action constraint effect on perceived distance is still highly debated (Firestone & Scholl, 2016). Some authors also suggested that action constraint effects reflect experimental demand bias on judgement rather than genuine perceptual effects (e.g., Durgin, Klein, Spiegel, Strawser & Williams, 2012).

OBJECTIVES

(1) Estimation of the **overall effect size** of action constraints on distance perception
(2) Estimation of two moderators (i.e., type of measure and experimental design) that could indicate that **experimental demand** plays a role in these effects
(3) Sample size recommendation according to the effect size of interest, for future replications

OVERALL EFFECT



33 outcomes extracted from 22 articles (N = 808), g = 0.47, 95% CrI [0.28, 0.67], $\tau_{article}$ = 0.36, 95% CrI [0.20, 0.56], τ_{studie} = 0.09, 95% CrI [0.01, 0.22], BF_{10} = 1333. The BF indicates very strong evidence for the evidence of the existence of the effect relative to its non-existence, the data are 1333 times more likely under the hypothesis of a non-null effect than under the hypothesis of a null effect.

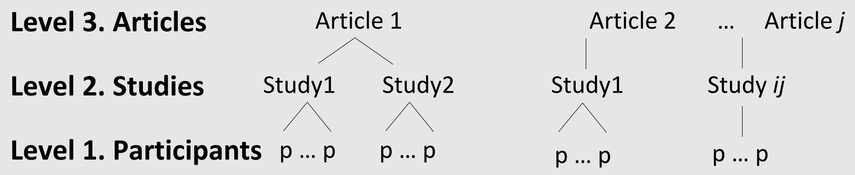
METHOD

INCLUSION CRITERIA

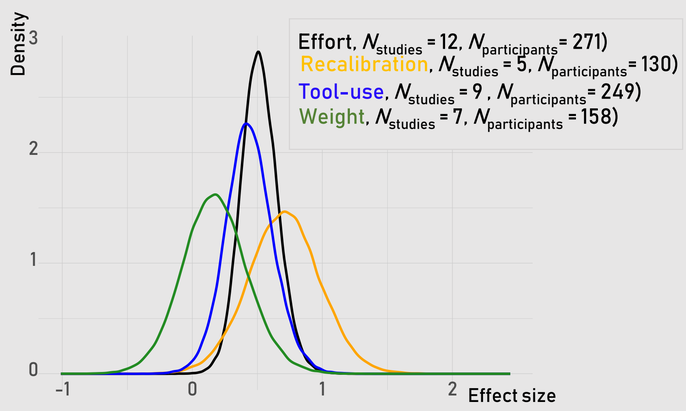
(1) Dependent variable = measure of distance estimation
(2) Independent variable = manipulation of an action constraint
(3) Distance estimation from an egocentric point of view

META-ANALYTIC MODEL

We used a three-levels meta-analytic Bayesian model to estimate the overall effect size, and standard deviation between articles ($\tau_{article}$) and between studies (τ_{studie}).



CONSTRAINT MANIPULATION



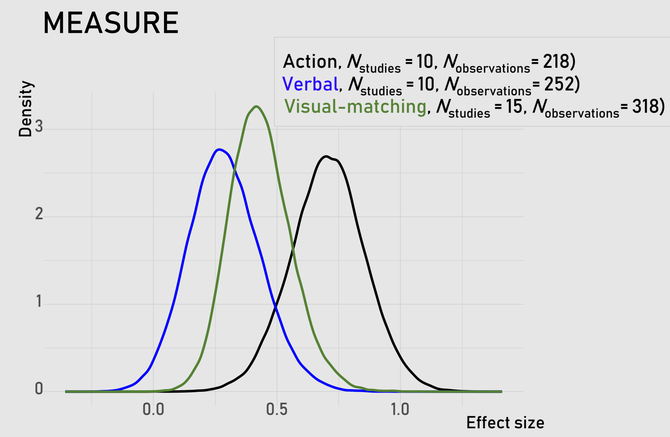
TOTAL SAMPLE SIZE RECOMMENDATION FOR POWER = .90

Constraint	g [95%CrI]	Between-subject	Within-subject
Effort	0.52 [0.25, 0.81]	157	41
Recalibration	0.7 [0.16,1.25]	88	23
Tool-use	0.44 [0.08, 0.82]	219	56
Weight	0.17 [-0.32,0.69]	1456	366

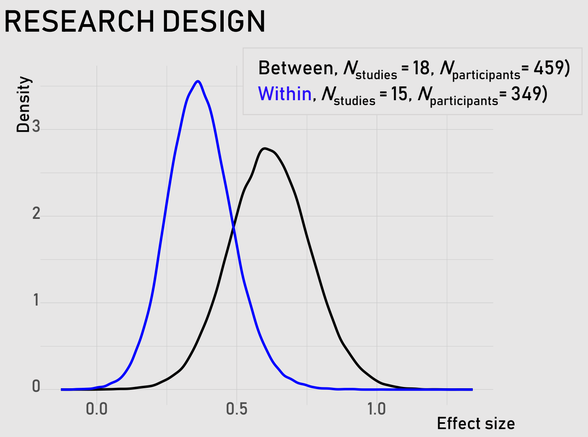
The only non-null manipulation is the weight (e.g., backpack, heavy ball throwing).

MODERATOR ANALYSES

MEASURE



RESEARCH DESIGN

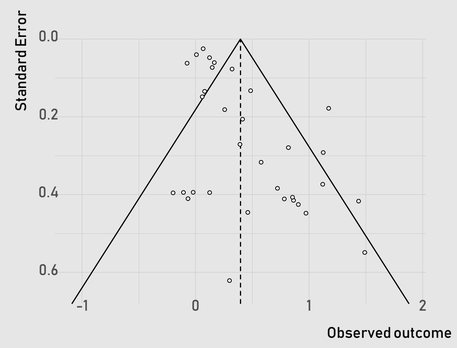


POSTERIOR DISTRIBUTION OF THE EFFECT SIZE

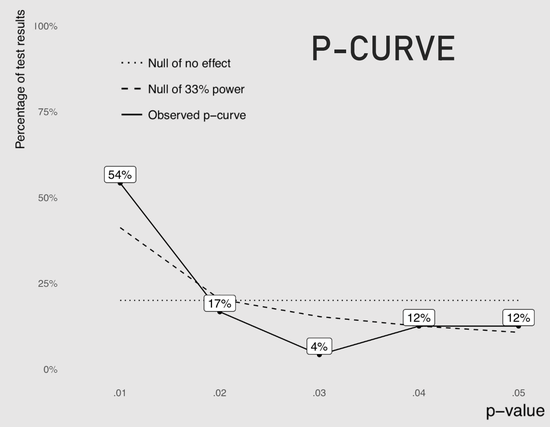
Within-subject designs and verbal measures are more likely to be biased by experimental demand (e.g., Hutchison & Loomis, 2006; Woods et al., 2009, Philbeck & Witt, 2015). Thus, if experimental demand inflates AC effects, they should be larger in these cases. However, verbal measures do not produce greater effect sizes than other measures, and within-subject design do not produce greater effect sizes than between-subject design.

ADDITIONAL ANALYSES

FUNNEL PLOT



P-CURVE



The funnel plot is roughly symmetrical as is the case of no publication bias, and the observed distribution of p-values (p-curve) from our database is close to a distribution that would be expected with a non-null effect.

CONCLUSIONS

Our meta-analysis indicates that people estimate a target farther away of 0.47 standard deviation unit when they are in high constraint condition than in low constraint condition, with a 95% probability of falling in the range from 0.28 to 0.67. However, the prevalence of published studies and the sample size of individual studies (~36 participants/study) is probably leading to an effect size overestimation. **For these reasons, we think future studies should focus on replicating AC effect with larger sample sizes.** Concerning experimental demand bias, our moderator analysis does not support the experimental demands hypothesis.

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